Activity Guide — middle / module-12-movement-of-sun

IKS Curriculum

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Activity 01 — Shadow-Stick (Gnomon) Observation

Module: Movement of the Sun · **Band:** Middle · **Day:** 1 · **Time:** 15 min (within Day 1’s 45 min)

Type

Outdoor / classroom-window observational science

Goal

Each pair measures and records the position of a gnomon’s shadow at two different times, demonstrating the Sun’s apparent daily motion.

Materials per pair

- One 30 cm wooden dowel, pencil, or chopstick (the **gnomon**)
- One A4 piece of stiff cardboard or chart paper (the **dial**)
- A small lump of clay or putty (to fix the gnomon upright)
- A compass (one per pair, or shared between two pairs)
- A ruler
- A pencil and a pen
- A stopwatch / phone timer for the second reading

Set-up (teacher, before class)

- Identify a sunny outdoor spot or an indoor location where direct sunlight enters through a window for at least 20 minutes during the lesson.
- Cut cardboard sheets in advance.
- Pre-mark a small N arrow on each cardboard (or instruct students to do this with the compass).

Instructions (student-facing — printable handout)

Shadow-stick observation — work in pairs.

Step 1 (3 min). Fix the gnomon upright on the cardboard with the clay. Use the compass to mark NORTH on the cardboard. Label the four cardinal directions (N, E, S, W).

Step 2 (1 min). Note the time on your stopwatch. Write it on the cardboard.

Step 3 (1 min). Trace the gnomon's shadow with a pencil. At the tip of the shadow, mark a small dot and write "T1" next to it.

Step 4 (10 min). WAIT — do not touch the gnomon or cardboard. While waiting, work in your notebook on the following questions: - Which direction is the shadow pointing? (Mostly opposite the Sun's position — so if Sun is in the east, shadow points west.) - Predict: in 10 minutes, will the shadow have moved CLOCKWISE or ANTICLOCKWISE around the gnomon? (Hint: as the Sun moves east-to-west, the shadow moves west-to-east — but it sweeps around the gnomon. From above the gnomon, the shadow rotates *clockwise* if you're in the Northern Hemisphere.)

Step 5 (1 min). At the 10-minute mark, trace the new shadow. Mark its tip with a "T2" dot.

Step 6 (5 min). Measure: how many degrees did the shadow rotate? (Use a protractor; the angle from T1 to T2 around the gnomon base.) - Expected: about **2.5°** in 10 minutes (since the Sun moves 360° in ~24 hours = 15°/hour = 2.5°/10 min).

Worksheet (one per pair)

Reading	Time	Shadow length	Shadow direction (which way it points from the gnomon)
T1		cm	
T2		cm	

Angle between T1 and T2 (around the gnomon base): ______ degrees

Predicted angle: 2.5° per 10 minutes. **Did your observation match?** ☐ Yes ☐ No (off by _____ °)

Reflection (one sentence): Which direction did the Sun move while you were watching? (Answer: from your earlier position to a later position — from east toward south toward west.)

Discussion prompts (last 3 min, whole class)

- Whose shadow rotated faster or slower than 2.5°? Why? (Errors in timing, sloppy tracing, gnomon not vertical.)
- If we did this on the December solstice instead of today, would the shadow be longer or shorter? (Answer: depends on time of year; longer in winter, shorter in summer at noon.)
- If we drew shadows at 9 AM, noon, and 3 PM on the same day, what shape would the three tips trace? (Answer: a curve — not a straight line. At the equinox, it's a straight line; at solstices, a curve that bends away from north.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners holding the cardboard steady, one tracing, one timing; quiet observation	“The shadow has moved!” / “It looks shorter than 10 minutes ago”
Needs intervention	Gnomon falling over; cardboard moved during the 10-min wait; one student looking at the Sun	Silence; or “I can’t see anything” because the cardboard was moved

Safety / sensitivity

- **CRITICAL: Never look at the Sun directly.** Watch the shadow. Anyone caught glancing at the Sun is sent indoors for the rest of the activity. This is non-negotiable.
- If the schoolyard is unsafe for outdoor work, do this at a sunny window.
- Wear hats if outdoors for more than 15 minutes. Sunscreen if available.

Differentiation

- **Tier up:** Predict the shadow length you’d expect from your gnomon (length × tan of the Sun’s zenith angle). The Sun’s zenith angle today (at noon) ≈ |latitude − solar declination|. Compute and compare with measurement.
- **Tier down:** Just observe — no measurement. Note: “shadow moved from here to here.”

Clean-up

- Take the cardboard back to the classroom — students keep their dials in their workbooks.
- Return the gnomons to the teacher.
- Compasses returned.

Activity 1 — Sunrise Direction Tracking

Module: Movement of the Sun · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Movement of the Sun through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

compass app, classroom window, notebook

What students do

Over 5 days, students note the compass bearing of sunrise each morning (from the same window). They graph the bearing and predict how it will change in 3 months.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Makara Saṁkrānti Explainer

Module: Movement of the Sun · **Band:** Middle · **Day:** 6 · **Time:** 15 min (within Day 6’s 45 min)

Type

Small-group prepared explanation; cross-cultural discussion

Goal

Each small group prepares and delivers a 2-minute explanation of *Makara Saṁkrānti* that distinguishes the **astronomy** (Sun’s transit into *Makara* rāśi) from the **regional festivals** that attach to it.

Materials per group

- One A4 sheet — the “Saṁkrānti explainer card”
- Coloured pencils / sketch pens
- Map of India (printed, one per group, with state outlines visible)
- Reference printout: list of regional festival names tied to Makara Saṁkrānti

Set-up

- Form groups of 3–4. Mix students from different family backgrounds where possible.

Instructions (student-facing — printable handout)

Makara Saṁkrānti explainer — work in your group.

Prepare a 2-minute explanation. Your group will present it to one neighbouring group.

Your explanation must answer **all five questions** below:

1. **What astronomical event is *Makara Saṁkrānti*?** *(Answer: the* saṁkrānti* — moment of transit — when the Sun enters the *rāśi* of *Makara* (Capricorn) in the *nirayana* / sidereal system. It happens around 14 January.)*
2. **What does it traditionally mark?** *(Answer: the start of* uttarāyaṇa* — the Sun's northward course — in the *nirayana* reckoning. It's traditionally considered auspicious in many Indian cultures.)*
3. **What regional festivals share this date?** *(At least four of: Pongal, Lohri, Bihu, Maghi, Uttarayan, Khichdi, Suggi, Magha Sankranti. Mark them on the map of India.)*
4. **What is one common practice across these festivals?** *(Likely answers: harvest celebration, kite-flying, sesame-jaggery sweets, bonfires. Take whichever you actually know; don't invent.)*
5. **What is the astronomical difference between Makara Saṁkrānti and the winter solstice?** *(Answer: the winter solstice is December 21 in the* sāyana* system. Makara Saṁkrānti is January 14 in the *nirayana* system. The ~24-day gap is the **ayanāṁśa** caused by precession over ~1700 years. The two were aligned ~1700 years ago.)*

Worksheet (the explainer card)

Question	Your group's answer (in 1–2 sentences)
1. What astronomy?	
2. What does it mark?	
3. Regional festivals	
4. Common practice	
5. Sāyana vs Nirayana gap	

Presentation order (within the group)

Decide:

- Who speaks Q1 and Q2 (the astronomy half)
- Who speaks Q3 and Q4 (the cultural half)
- Who speaks Q5 (the comparison)
- Who hands the card to the listening group at the end

Pair-share format (last 5 min)

- Two groups face each other.

- Group A presents (2 min).
- Group B asks one question and offers one positive comment.
- Swap roles.

Discussion prompts (whole class, last 3 min)

- Did anyone present a festival you'd never heard of?
- Did any group accidentally describe the festival without describing the astronomy? (This is a common mistake. Acknowledge gently.)
- The 24-day gap is real and important. Without it, *Makara Saṁkrānti* would BE the winter solstice. Why isn't it?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Group huddled, distributing speaking parts, marking the map	"I'll take Q3 since I know Pongal" / "Let me read out Q5"
Needs intervention	One student dominating; group treating it as a festival presentation only	"It's just a festival, that's all" — redirect: "Yes, but WHAT astronomical event triggers the festival?"

Safety / sensitivity

- Some students may feel personally attached to one of these festivals; others may not celebrate any. Make clear: we are not ranking festivals or implying that one region's celebration is more "Indian" than another's. Each is rooted in the same astronomical event.

Differentiation

- **Tier up:** Group adds a *Karka Saṁkrānti* equivalent — what astronomically? What festivals? (Hint: there's no large pan-Indian festival on July 16 the way there is on January 14. Why might that be?)
- **Tier down:** Skip Q5. Answer Q1–Q4 only. The *ayanāṁśa* question can be revisited individually.

Clean-up

- Hand the explainer card to the teacher.
- Maps go back in the materials box.

Activity 03 — Sāyana vs Nirayana Compare-Contrast

Module: Movement of the Sun · **Band:** Middle · **Day:** 7 · **Time:** 10 min (within Day 7's 45 min)

Type

Paired written compare-contrast

Goal

Each pair fills a two-column table comparing the *sāyana* (tropical) and *nirayana* (sidereal) zodiacs across six dimensions, demonstrating clear understanding of why they differ.

Materials per pair

- One printed worksheet (next page)
- A pen
- The Day 7 board diagram (or notebook copy of it)

Set-up

- Distribute one worksheet per pair. Pairs can be the same as Days 5–6 working pairs.

Worksheet — Sāyana vs Nirayana

Pair name: _______________ & _______________ Date: __________

Fill in the table. Each cell should be one sentence.

Question	*Sāyana* (tropical)	*Nirayana* (sidereal)
1. The Sun enters *Meṣa* (Aries) on what date?		
2. The Sun enters *Makara* (Capricorn) on what date?		
3. Fixed to what reference?		
4. What changes over centuries?		
5. Used by modern astronomy (Y/N)?		
6. Used by traditional Indian *pañcāṅga* (Y/N)?		

Short answer (3–4 sentences below the table):

In your own words, explain why both systems are “right” for their own purposes. Use the term *ayanāṁśa* somewhere in your answer.

Answer key (teacher copy)

Question	*Sāyana*	*Nirayana*
1. Sun enters Meṣa	~21 March (the spring equinox by definition)	~14 April (when the Sun crosses the sidereal *Meṣa* boundary)
2. Sun enters Makara	~21 December (the winter solstice by definition)	~14 January (Makara Saṁkrānti)
3. Fixed to what?	The equinoxes (the points where Sun crosses celestial equator)	A specific point against the fixed stars (chosen near *Citrā* / Spica)
4. What changes?	The position of the equinoxes against the stars (precession)	The alignment with the seasons (the same precession, seen the other way around)
5. Modern astronomy uses?	Yes — civil calendars, weather science, Western horoscopes	No (modern astronomy uses *sāyana* for civil dates; uses pure star coordinates for catalogues, but not the *rāśi* boundaries)
6. Indian *pañcāṅga* uses?	No (almost universally)	Yes

Short-answer rubric: the answer should include:

- The point that *neither* is right — they're answering different questions.
- The point that the *seasons* and the *stars* drift relative to each other because of precession.
- The word *ayanāṁśa* used in context (the offset, ~24° today).
- A consequence: e.g. "Western horoscopes use *sāyana*, Indian *pañcāṅgas* use *nirayana*, so a person's 'Sun sign' may be one *rāśi* different in the two systems."

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair pointing at the Day 7 diagram, debating one cell at a time	"Wait — is *Makara* on Dec 21 or Jan 14?" / "Both, in different systems!"
Needs intervention	Both cells filled with the same answer; or one student copying from the board without discussion	Silence; one-word answers

Discussion prompts (whole-class, last 3 min if time)

- Did any pair disagree on Q5 (modern astronomy) or Q6 (*pañcāṅga*)? Why?

- A challenge: if Earth's axis suddenly stopped precessing tomorrow, how long would it take for *Makara Saṁkrānti* to drift to coincide with the winter solstice? *(Trick question: it wouldn't drift back — it would just stay 24° offset forever. The drift CAUSED the gap, and stopping precession would just freeze the gap.)*

Differentiation

- **Tier up:** Compute the year in which the two systems WILL re-align by precession. Hint: ~25,800 years − 1700 years = ~24,100 years from now.
- **Tier down:** Fill only Q1, Q2, Q5, Q6. The "what changes" and the short-answer can be skipped.

Project Brief

Project Brief — Personal Solar Calendar

Module: Movement of the Sun · **Band:** Middle

What you will do

Make a one-page solar calendar for the current year: mark the two solstices, two equinoxes, all 12 saṁkrānti dates, plus your own birthday with its rāśi.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (300 words (Middle)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.

- **Day 10:** share day.

Sources

You must cite at least 2 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.